Chenhuan Cai (Choy)

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EDUCATION

University of Zurich Sep 2023 - Sep 2026

M.Sc. in Artificial Intelligence Zurich

- Foundations of Data Science (FDS) Assistant for the fall semester of 2024
- GPA: 5.25 / 6.00. Core courses: Introduction to Machine Learning, Deep Learning, Computer Vision, Digital Humans, etc.

Harbin Institute of Technology (Shenzhen) Sep 2019 - Jun 2023

B.Eng. in Big Data Technology and Data Science Shenzhen

- GPA: 3.7 / 4.0. Core courses: Data Structure, Advanced Algebra, Probability Theory, Statistics, Complex Functions, Differential Equations, Real Analysis, Statistical Machine Learning, etc.
- First Prize of Academic Scholarship (Top 5%), First Prize of Mathematics Competition for College Students (Top 8%)

RESEARCH & INTERNSHIP

ETH Zurich May 2025 - Dec 2025

3D Vision Research, Computer Vision Group (CVG) Zurich

- Sparse-view 3D Gaussian reconstruction.
- Proposed ***GenSplat***, a generative-prior-based Gaussian fusion framework with more robust Gaussian generation.
- Generative prior: leveraged diffusion models and camera pose selection strategies for data augmentation to improve quality.
- Gaussian fusion: applied a reference-free Transformer and local-to-global Gaussian fusion to support multi-view and remove input-order dependency. Achieved better quantitive and qualitative results than NoPoSplat on RE10K and DL3DV.

NetEase Youdao Information Technology (Beijing) Co., Ltd Apr 2022 - Jun 2022

Computer Vision Algorithm Engineer Beijing

- Developed vision algorithms for hand action detection in Youdao Aladdin smart desk lamp.
- Phase 1: explored pseudo-labeling and image processing, evaluated image classification and YOLO variants, and delivered a YOLOX solution for successful detection when the lamp is on.
- Phase 2: expanded the dataset and applied data augmentation and tracking methods to handle lamp-off scenarios, reaching 80%+ accuracy, and verified feasibility for production deployment.

PUBLICATION

STaR: Multi-Granular Spatio-Temporal Reasoning for Long-Form DVC. ECAI 2025. Feb 2024 - Aug 2024

Wu, Y.*, Cai, C.*, Li, H., Bai, Z. & Yan, L. (* Equal contribution.)

- Cross-modal fusion for long-form dense video captioning; achieved SOTA on SoccerNet-Caption.
- Proposed ***STaR***, a multi-granular spatio-temporal reasoning framework that models both local and global visual context and incorporates dynamic positional encoding for cross-view spatial semantic modeling.
- Designed a Context Capture (CC) module to fuse global context with short-window local features, and a Spatial Semantic Synthesis (SSS) module with hash encoding to enhance spatial perception.

ShapeNav: 3D Geometric Perception and Reasoning for Multi-Floor VLN. Under Review. Sep 2024 - May 2025

Wu, Y.*, Cai, C.*, Yan, L., Li, M., Xu, C., Yang, C., Guan, F. & Li, P. (* Equal contribution.)

- Incorporating explicit 3D geometry into vision-and-language navigation; achieved SOTA results on R2R and F2F.
- Proposed ***ShapeNav***, a point-cloud + VLM framework for 3D geometric perception and spatial understanding navigation, without requiring 3D pretraining.
- Designed Geometric Patch Extraction (GPE) and Masked Optimal Transport (MOT) to align 3D shape features with 2D visual features for cross-modal fusion.

COMPETITIONS REWARDS

Kaggle Competition: "Stable Diffusion – Image to Prompts" | Silver Medal. Feb 2023 - May 2023

- Task: Predict the corresponding prompt given a generated image.
- Built a large-scale training dataset from Stable Diffusion–generated images and used contrastive learning to align image and text embeddings, enabling accurate prompt reconstruction from images.

China Undergraduate Mathematical Contest in Modeling (CUMCM) | Second Prize Sep 2021

- Task: Analyze and model a chemical experimental dataset using machine learning methods.
- Primarily used correlation analysis methods, machine learning models (Random Forest, XGBoost, stacking ensembles), SHAP for interpretability, and optimization techniques (PSO).

MISCELLANEOUS

- **Research Interests:** Machine Learning, Image Recognition, Object Detection, Video Understanding, 3D Reconstruction, Gen erative Models, Multimodal Large Models, Multimodal Alignment
- **Skills:** Python, Linux, Git, Bash, Conda, CUDA, PyTorch, Hugging Face, Transformers/Diffusers, HPC/Slurm, etc.
- **Languages:** Chinese (native); English (proficient)